AGENTIC AI · CUSTOMER SUPPORT

Multi-Agent Customer Support

A **supervisor agent** plans each customer request and hands the work to order, policy and refund specialists. Large refunds wait for a person to approve.

LangGraph | Agentic RAG | Guardrails | Human-in-the-loop | Live demo

Presented by Lavanya Srivastava · Lavanya AI Consultancy

THE PROBLEM

Support teams answer the same hard questions all day

Slow answers

An agent checks the order system, then the policy documents, then finance, before the customer gets a reply.

Inconsistent policy

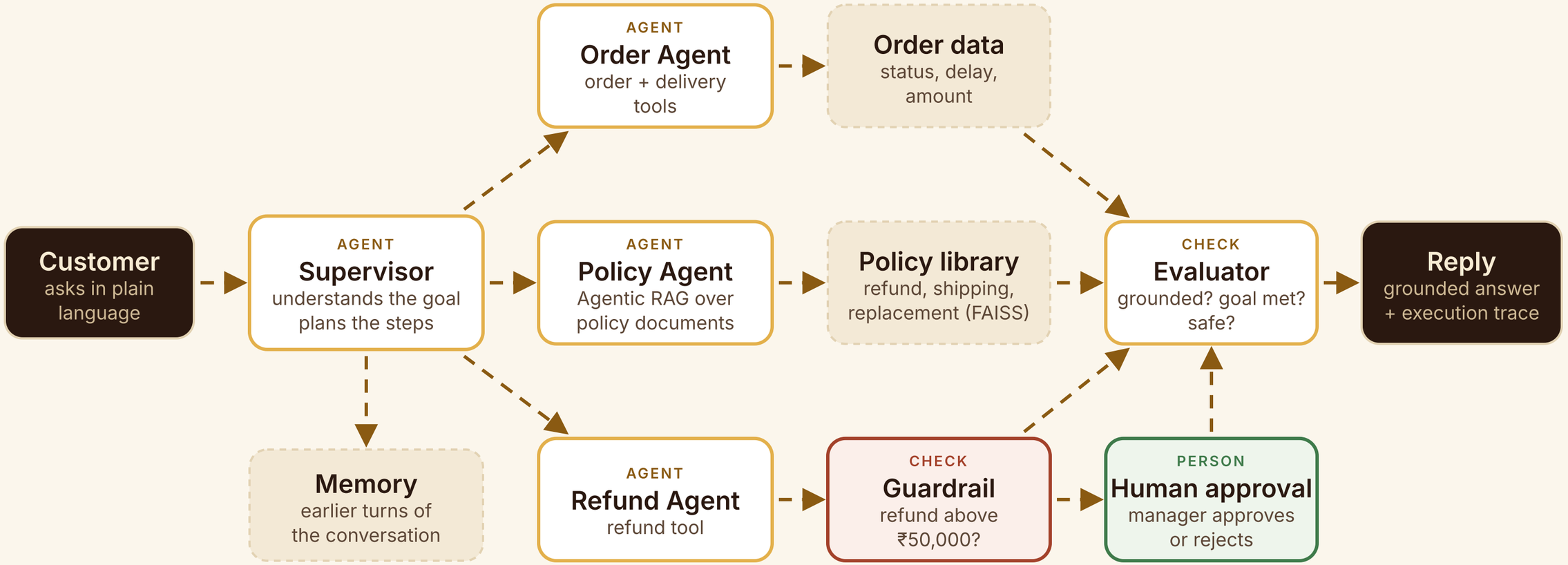
Different people read the refund policy differently, so customers get different answers.

Risky automation

A simple chatbot that can issue refunds could pay out large sums with nobody checking.

ARCHITECTURE

How one request flows through the agents



Supervisor → specialists → tools → guardrail → person → evaluation. One request, a safe and traceable outcome.

WHAT MAKES IT ENTERPRISE-READY

More than a chatbot

Planning

The supervisor turns each goal into an operational plan and tracks it to completion or a safe pause.

Specialist agents

Orders, policy and refunds each have a dedicated agent with its own tools.

Agentic RAG

Policy answers come from the company's documents, not the model's memory.

Memory

Follow-up questions keep their context across the conversation.

Guardrails + people

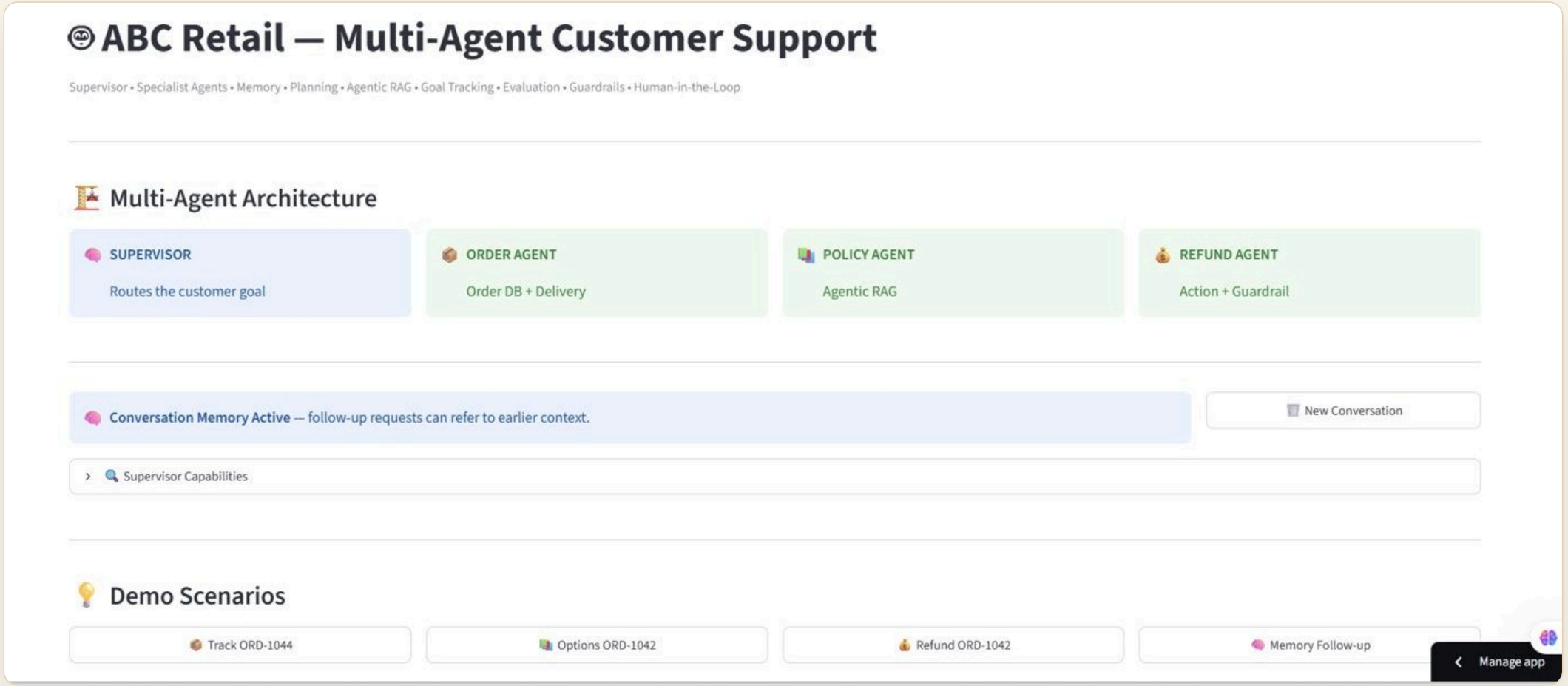
Money above a set limit always needs a human decision.

Evaluation + trace

Every workflow is checked and fully visible for audit.

SEE IT WORKING

The live application



Architecture panel, conversation memory and one-click demo scenarios: track an order, see options, request a refund, ask a follow-up.

RULES AND TECHNOLOGY

Built on clear business rules

₹50,000
automatic refund limit; above it a person approves

7 days
delivery delay that makes an order eligible

4 agents
supervisor plus order, policy and refund specialists

Layer	Technology
Orchestration	LangGraph + LangChain
Language model	OpenAI
Policy retrieval	FAISS + sentence-transformers
Interface and quality	Streamlit · unit tests for tools and limits, run in CI

NEXT STEP

A supervised AI agent for your support desk

The same pattern fits banking, telecom, e-commerce and student services: a specialist for each area, clear limits, and a person for the decisions that matter.

Open the live demo | View the code | Talk to Lavanya

Lavanya Srivastava · Lavanya AI Consultancy
lavanya.sriva@gmail.com · lavanya-portfolio-azure.vercel.app